

Right Triangle Trigonometry: Ratios, Triples, and Solving Triangles

Summary

This lecture introduces the foundational concepts of right triangle trigonometry using the SOH CAH TOA framework. Students learn how to identify sides relative to a reference angle, evaluate all six trigonometric functions (sine, cosine, tangent, cosecant, secant, and cotangent), and leverage Pythagorean triples to rapidly determine missing side lengths. Furthermore, the lesson covers practical algebraic methods for computing missing sides with primary trigonometric ratios and calculating unknown angles using inverse trigonometric functions.

Learning Objectives

- Identify the hypotenuse, opposite, and adjacent sides of a right triangle relative to a specified reference angle.
- Define and calculate the six trigonometric functions: sine, cosine, tangent, cosecant, secant, and cotangent.
- Recall and apply common Pythagorean triples and their scalar multiples to solve for missing side lengths.
- Set up algebraic equations using trigonometric ratios to solve for missing side lengths using a calculator in degree mode.
- Apply inverse trigonometric functions (arcsin, arccos, arctan) to calculate unknown acute angles from known side lengths.

Lecture Sections

Anatomy of a Right Triangle and the Six Trigonometric Ratios

Right triangle trigonometry is established upon the relationship between the acute angles of a right triangle and ratios of its sides. In any right triangle, the side located directly opposite the 90-degree right angle (often symbolized by a small square) is called the hypotenuse. The hypotenuse is always the longest side of the triangle. The designations of the remaining two sides depend entirely on the reference angle, θ . The side situated directly across from angle θ is the opposite side, while the side that touches angle θ and runs between it and the 90-degree angle is the adjacent side. To memorize the primary ratios, the mnemonic SOH CAH TOA is used: Sine is Opposite over Hypotenuse (SOH), Cosine is Adjacent over Hypotenuse (CAH), and Tangent is Opposite over Adjacent (TOA). In addition to these primary functions, there are three reciprocal trigonometric

functions: cosecant (csc), secant (sec), and cotangent (cot). Cosecant is the reciprocal of sine, defined as hypotenuse over opposite. Secant is the reciprocal of cosine, defined as hypotenuse over adjacent. Cotangent is the reciprocal of tangent, defined as adjacent over opposite. Whenever the three primary ratios are established for an angle, the reciprocal ratios can be directly obtained simply by inverting each fraction.

Examples

Reciprocal relationship demonstration

If a right triangle configuration produces a sine ratio of $\sin(\theta) = 4/5$, its reciprocal function cosecant is evaluated by flipping the fraction to yield $\csc(\theta) = 5/4$. Similarly, a cosine of $3/5$ flips to a secant of $5/3$, and a tangent of $4/3$ flips to a cotangent of $3/4$.

Definitions

Hypotenuse: The longest side of a right triangle, situated directly across from the 90-degree right angle.

Opposite Side: The side of a right triangle positioned directly across from the chosen reference angle theta (θ).

Adjacent Side: The side of a right triangle located next to the reference angle theta (θ), forming one of the rays of that angle alongside the hypotenuse.

SOH CAH TOA: A mnemonic device representing the three primary trigonometric ratios: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, and Tangent = Opposite/Adjacent.

Cosecant (csc): The reciprocal trigonometric function of sine, calculated as Hypotenuse / Opposite.

Secant (sec): The reciprocal trigonometric function of cosine, calculated as Hypotenuse / Adjacent.

Cotangent (cot): The reciprocal trigonometric function of tangent, calculated as Adjacent / Opposite.

Formulas

$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

$$\sec(\theta) = \frac{1}{\cos(\theta)} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

$$\cot(\theta) = \frac{1}{\tan(\theta)} = \frac{\text{Adjacent}}{\text{Opposite}}$$

Instructor Emphasis

- The hypotenuse is always the longest side and always sits opposite the right-angle box.
- Labels for opposite and adjacent must be determined relative to the angle theta being evaluated; swapping the reference angle swaps the opposite and adjacent labels.

- To find cosecant, secant, and cotangent, evaluate the primary ratios first and invert the fractions.

Common Misconceptions

- Confusing the adjacent side with the hypotenuse, since both sides touch the angle theta.
- Assuming cosecant corresponds to cosine because both start with 'co'; cosecant is the reciprocal of sine, whereas secant is the reciprocal of cosine.

Pythagorean Theorem, Pythagorean Triples, and Full Triangle Evaluation

To evaluate all six trigonometric ratios of a right triangle, all three side lengths must be known. When only two sides are given, the missing third side can be determined using the Pythagorean theorem, $a^2 + b^2 = c^2$, where c represents the hypotenuse and a and b represent the legs. Although the Pythagorean theorem can always solve for a missing side, recognizing Pythagorean triples provides an efficient shortcut for calculations. A Pythagorean triple consists of three positive integers that satisfy the equation $a^2 + b^2 = c^2$. The most frequent primary triples include (3, 4, 5), (5, 12, 13), (8, 15, 17), and (7, 24, 25). Any integer multiple of these base triples also forms a valid right triangle. For instance, multiplying (3, 4, 5) by 2 gives (6, 8, 10), multiplying by 3 gives (9, 12, 15), and multiplying by 5 gives (15, 20, 25). Less common triples that occasionally appear include (9, 40, 41) and (11, 60, 61). Once all three sides are known, the six trigonometric functions can be written out as exact rational numbers and simplified to their lowest fractional terms.

Examples

Evaluating the (3, 4, 5) base triangle

Given legs 3 and 4 with angle theta opposite side 4: $c^2 = 3^2 + 4^2 = 9 + 16 = 25$, so $c = 5$. The six trigonometric values are: $\sin(\theta) = 4/5$, $\cos(\theta) = 3/5$, $\tan(\theta) = 4/3$, $\csc(\theta) = 5/4$, $\sec(\theta) = 5/3$, and $\cot(\theta) = 3/4$.

Evaluating the (8, 15, 17) triangle

Given adjacent leg 8 and hypotenuse 17 relative to theta: $8^2 + b^2 = 17^2$ implies $64 + b^2 = 289$, so $b^2 = 225$ and $b = 15$. The ratios are: $\sin(\theta) = 15/17$, $\cos(\theta) = 8/17$, $\tan(\theta) = 15/8$, $\csc(\theta) = 17/15$, $\sec(\theta) = 17/8$, and $\cot(\theta) = 8/15$.

Recognizing a scaled triple (15, 20, 25)

Given hypotenuse 25 and adjacent side 15: recognizing these as 5×5 and 3×5 indicates a 3-4-5 scaled triple. The missing opposite side is $4 \times 5 = 20$. The ratios simplify as follows: $\sin(\theta) = 20/25 = 4/5$, $\cos(\theta) = 15/25 = 3/5$, $\tan(\theta) = 20/15 = 4/3$, $\csc(\theta) = 5/4$, $\sec(\theta) = 5/3$, and $\cot(\theta) = 3/4$.

Definitions

Pythagorean Theorem: A fundamental geometric principle stating that in a right triangle with legs a and b and hypotenuse c , the sum of the squares of the legs equals the square of the hypotenuse ($a^2 + b^2 = c^2$).

Pythagorean Triple: A set of three positive integers (a , b , c) that perfectly satisfy the Pythagorean theorem $a^2 + b^2 = c^2$.

Formulas

$$a^2 + b^2 = c^2$$

$$c = \sqrt{a^2 + b^2}$$

$$b = \sqrt{c^2 - a^2}$$

Instructor Emphasis

- Memorize the key triples: (3, 4, 5), (5, 12, 13), (8, 15, 17), and (7, 24, 25).
- Remember that whole number multiples of triples (such as 6-8-10 or 15-20-25) are valid right triangles and save substantial calculation time.
- Always simplify trigonometric ratios to lowest terms (e.g., 20/25 simplifies to 4/5).

Common Misconceptions

- Applying the Pythagorean theorem with the hypotenuse substituted into a or b rather than c.
- Forgetting to take the square root of c^2 when solving for the final side length.

Solving for Missing Sides Using Trigonometric Equations

When an acute angle and one side of a right triangle are known, standard algebraic manipulation paired with primary trigonometric ratios can find either of the remaining unknown sides. The process starts by cataloging the given information relative to the known acute angle. You identify which side is provided (opposite, adjacent, or hypotenuse) and which side represents the target variable x . Based on this pair, choose the corresponding ratio: use sine for opposite and hypotenuse, cosine for adjacent and hypotenuse, or tangent for opposite and adjacent. Once the trigonometric equation is established, isolate x algebraically. If x is in the numerator (for instance, $\tan(38^\circ) = x/42$), multiply both sides by the denominator to yield $x = 42 \cdot \tan(38^\circ)$. If x is in the denominator (for instance, $\sin(32^\circ) = 12/x$), cross-multiply to reach $x \cdot \sin(32^\circ) = 12$, then divide both sides by the trigonometric term to get $x = 12 / \sin(32^\circ)$. A critical technical prerequisite is verifying that the scientific or graphing calculator is explicitly configured in degree mode; computing angles in radian mode produces incorrect numerical values.

Examples

Solving for opposite side x with given adjacent side

Given angle 38° and adjacent side 42: set up $\tan(38^\circ) = x / 42$. Multiply both sides by 42: $x = 42 \cdot \tan(38^\circ)$. Since $\tan(38^\circ) \approx 0.7813$, $x \approx 42 \cdot 0.7813 \approx 32.8$.

Solving for adjacent side x with given hypotenuse

Given angle 54° and hypotenuse 26: set up $\cos(54^\circ) = x / 26$. Multiply both sides by 26: $x = 26 \cdot \cos(54^\circ)$. Since $\cos(54^\circ) \approx 0.5878$, $x \approx 26 \cdot 0.5878 \approx 15.28$.

Solving for hypotenuse x in the denominator

Given angle 32° and opposite side 12: set up $\sin(32^\circ) = 12 / x$. Cross-multiply: $x \cdot \sin(32^\circ) = 12$. Divide by $\sin(32^\circ)$: $x = 12 / \sin(32^\circ)$. Evaluating on a calculator yields $x \approx 22.64$.

Definitions

Degree Mode: A setting on scientific and graphing calculators ensuring trigonometric functions interpret input values and outputs in degrees rather than radians.

Formulas

$$x = \text{Adjacent} \cdot \tan(\theta)$$

$$x = \text{Hypotenuse} \cdot \cos(\theta)$$

$$x = \frac{\text{Opposite}}{\sin(\theta)}$$

Instructor Emphasis

- Always confirm that the calculator is set to degree mode before entering calculations.
- When the variable x is in the denominator, cross-multiply first to avoid algebraic errors.

Common Misconceptions

- Leaving the calculator in radian mode, which results in completely wrong numerical answers.
- Multiplying instead of dividing when the unknown variable x resides in the denominator of the ratio.

Finding Missing Angles Using Inverse Trigonometric Functions

When two sides of a right triangle are known but an acute angle θ is unknown, inverse trigonometric functions (also known as arc functions: $\arcsin$, $\arccos$, and $\arctan$) are used to solve for the angle. While standard trigonometric functions accept an angle measure as an input and output a side ratio, inverse trigonometric functions accept a side ratio as an input and return the corresponding angle measure. To find an unknown angle θ , first identify the relationship of the two known side lengths to θ (opposite, adjacent, or hypotenuse). Formulate the corresponding primary trigonometric ratio. Then, isolate θ by applying the appropriate inverse operation to both sides of the equation. For example, if $\tan(\theta) = 5/4$, applying the inverse tangent gives $\theta = \arctan(5/4)$ or $\tan^{-1}(5/4)$. Entering this into a calculator in degree mode yields the measure of θ in degrees. This same inverse technique applies across all three primary relationships whenever two side measurements are accessible.

Examples

Finding an angle using inverse tangent ($\arctan$)

Given opposite side 5 and adjacent side 4: write $\tan(\theta) = 5/4$. Take the inverse tangent: $\theta = \arctan(5/4)$. Entering $\arctan(1.25)$ into the calculator produces $\theta \approx 51.34^\circ$.

Finding an angle using inverse cosine ($\arccos$)

Given adjacent side 3 and hypotenuse 7: write $\cos(\theta) = 3/7$. Take the inverse cosine: $\theta = \arccos(3/7)$. Entering $\arccos(3/7)$ into the calculator produces $\theta \approx 64.62^\circ$.

Finding an angle using inverse sine (arcsin)

Given opposite side 5 and hypotenuse 6: write $\sin(\theta) = 5/6$. Take the inverse sine: $\theta = \arcsin(5/6)$. Entering $\arcsin(5/6)$ into the calculator produces $\theta \approx 56.44^\circ$.

Definitions

Inverse Trigonometric Function: Mathematical operations ($\arcsin$, $\arccos$, $\arctan$) that reverse the action of trigonometric ratios, accepting a side ratio as input to return the corresponding angle measure.

Formulas

$$\theta = \arctan\left(\frac{\text{Opposite}}{\text{Adjacent}}\right) = \tan^{-1}\left(\frac{\text{Opposite}}{\text{Adjacent}}\right)$$

$$\theta = \arccos\left(\frac{\text{Adjacent}}{\text{Hypotenuse}}\right) = \cos^{-1}\left(\frac{\text{Adjacent}}{\text{Hypotenuse}}\right)$$

$$\theta = \arcsin\left(\frac{\text{Opposite}}{\text{Hypotenuse}}\right) = \sin^{-1}\left(\frac{\text{Opposite}}{\text{Hypotenuse}}\right)$$

Instructor Emphasis

- A calculator is necessary to evaluate inverse trigonometric values for non-special angle ratios.
- Ensure the calculator is set to degree mode to avoid getting angles output in radians.

Common Misconceptions

- Confusing inverse notation $\tan^{-1}(x)$ with the reciprocal $1/\tan(x)$ (cotangent). The notation $\tan^{-1}$ denotes an inverse function, not an algebraic exponent of -1.

Questions & Answers

Q: What does the expression SOH CAH TOA mean?

A: SOH CAH TOA is a mnemonic for the three primary trigonometric ratios: SOH stands for Sine = Opposite / Hypotenuse, CAH stands for Cosine = Adjacent / Hypotenuse, and TOA stands for Tangent = Opposite / Adjacent.

Q: How are cosecant, secant, and cotangent obtained from sine, cosine, and tangent?

A: They are reciprocal functions: cosecant is $1 / \text{sine}$ (Hypotenuse / Opposite), secant is $1 / \text{cosine}$ (Hypotenuse / Adjacent), and cotangent is $1 / \text{tangent}$ (Adjacent / Opposite). Simply flip the fraction of the corresponding primary ratio.

Q: What are some common Pythagorean triples mentioned in the lecture?

A: Common triples include (3, 4, 5), (5, 12, 13), (8, 15, 17), and (7, 24, 25), as well as any scalar multiples of these sets (e.g., 6-8-10 or 15-20-25). Less common triples include (9, 40, 41) and (11,

60, 61).

Q: What algebraic step should be taken when the unknown variable x is in the denominator of a trigonometric equation (e.g., $\sin(32^\circ) = 12 / x$)?

A: Cross-multiply to clear the fraction, resulting in $x \cdot \sin(32^\circ) = 12$, then divide both sides by $\sin(32^\circ)$ to solve for x ($x = 12 / \sin(32^\circ)$).

Q: How do you calculate an unknown acute angle when two side lengths of a right triangle are known?

A: Determine which trigonometric ratio relates the two known sides to the angle, then evaluate the inverse of that trigonometric function (arcsin, arccos, or arctan) of the ratio on a calculator set to degree mode.

Revision Summary

Right triangle trigonometry relies on identifying sides relative to reference angle θ : the hypotenuse is opposite the 90° angle, while opposite and adjacent sides are defined by θ . SOH CAH TOA establishes the primary ratios: $\sin(\theta) = \text{Opp}/\text{Hyp}$, $\cos(\theta) = \text{Adj}/\text{Hyp}$, and $\tan(\theta) = \text{Opp}/\text{Adj}$. Reciprocal ratios are formed by inverting these fractions: $\csc(\theta) = \text{Hyp}/\text{Opp}$, $\sec(\theta) = \text{Hyp}/\text{Adj}$, and $\cot(\theta) = \text{Adj}/\text{Opp}$. Knowing standard Pythagorean triples—such as (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25), and their multiples—allows rapid computation of unknown sides without manual square root calculations. To find an unknown side length, set up the primary trigonometric ratio matching the known side and unknown variable, cross-multiplying or isolating x algebraically while ensuring the calculator is in degree mode. To find an unknown angle from two known sides, set up the appropriate ratio and apply its inverse function (arcsin, arccos, or arctan).